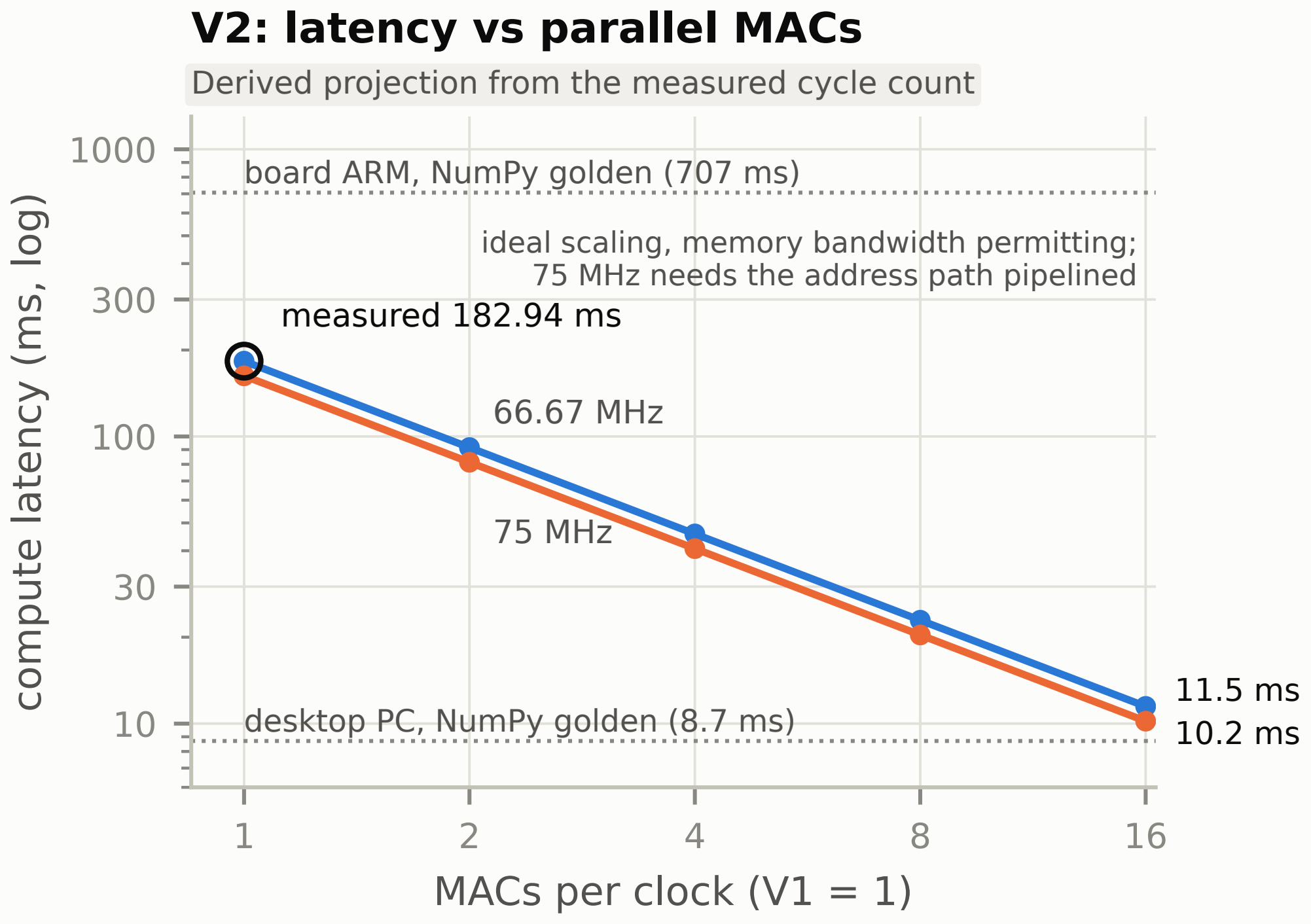
V2: latency vs parallel MACs
Derived projection from the measured cycle count
compute latency (ms, log)
1000
300
100
30
10
board ARM, NumPy golden (707 ms)
ideal scaling, memory bandwidth permitting;
75 MHz needs the address path pipelined
measured 182.94 ms
66.67 MHz
75 MHz
11.5 ms
10.2 ms
desktop PC, NumPy golden (8.7 ms)
1
2
4
8
16
MACs per clock (V1 = 1)